Encryption Match-Up

Instructor key · Req 6a · 5 minutes in the Encryption block

What it teaches

Encryption turns readable information into scrambled text that only the right key can unlock. Wardens match each use to the reason it matters, then name the type of encryption.

Before class

Print page 2 once per lodge (6 copies, or 7 for a spare) and cut apart. Each lodge gets three piles: USES (green), REASONS (white) and TYPES (yellow). Shuffle each pile. The slide shows four matches. The fifth, Saved passwords, is a spare for fast lodges.

Run it

1) Lodges match each USE to its REASON for 2 minutes, then add the TYPE. 2) Click through the slide. Each click reveals one correct match with its type. Lodges check theirs. 3) Ask why each type matters. **Points:** +1 for each correct pair and type of encryption.

#	Use	Reason	Type	Say
1	**Online banking**	Keeps your account details private	**HTTPS / TLS**	The padlock and https in the address bar scramble what you send to the bank.
2	**Messaging app**	Only sender and receiver can read	**End-to-end encryption**	Only the two phones hold the keys, so not even the app company can read the message.
3	**Home Wi-Fi**	Keeps neighbors off your network	**WPA2 / WPA3**	Use a strong Wi-Fi password and a modern security setting on the router.
4	**Locked phone**	Protects photos if the phone is lost	**Device encryption**	A phone with a passcode scrambles what is stored on it until it is unlocked.
5	**Saved passwords**	Protects your accounts if your device is stolen	**Encrypted password vault**	A password manager keeps passwords scrambled behind one strong master password.

Note: https tells you the connection is encrypted, not that the site is trustworthy. A phishing site can use https too.

Sealed Scroll match-up cards

Cut apart · one set per lodge

USES	REASONS	TYPES
Online banking	Protects your accounts if your device is stolen	HTTPS / TLS
Messaging app	Keeps your account details private	End-to-end encryption
Home Wi-Fi	Protects photos if the phone is lost	WPA2 / WPA3
Locked phone	Only sender and receiver can read	Device encryption
Saved passwords	Keeps neighbors off your network	Encrypted password vault